DISCUSSION PAPER SERIES

DP22123

TAX, LIES AND RED TAPE

Megha Patnaik

DEVELOPMENT ECONOMICS AND
PUBLIC ECONOMICS



ISSN 0265-8003

TAX, LIES AND RED TAPE

Megha Patnaik

Discussion Paper DP22123
Published 03 November 2026
Submitted 30 October 2026

Centre for Economic Policy Research
187 boulevard Saint-Germain, 75007 Paris, France
2 Coldbath Square, London EC1R 5HL
Tel: +44 (0)20 7183 8801
www.cepr.org

This Discussion Paper is issued under the auspices of the Centre's research programmes:

- Development Economics
- Public Economics

Any opinions expressed here are those of the author(s) and not those of the Centre for Economic Policy Research. Research disseminated by CEPR may include views on policy, but the Centre itself takes no institutional policy positions.

The Centre for Economic Policy Research was established in 1983 as an educational charity, to promote independent analysis and public discussion of open economies and the relations among them. It is pluralist and non-partisan, bringing economic research to bear on the analysis of medium- and long-run policy questions.

These Discussion Papers often represent preliminary or incomplete work, circulated to encourage discussion and comment. Citation and use of such a paper should take account of its provisional character.

Copyright: Megha Patnaik

TAX, LIES AND RED TAPE

Abstract

Do family firms evade tax when enforcement is weak? I combine a measure of ultimate ownership in India with firm-level excise tax payments and statutory rates. Firms with greater family shareholding have lower effective tax rates. To identify differences in compliance, I use year-by-year changes in statutory rates across products. A one-percentage-point increase in the statutory rate for a firm raises the effective rate by 0.14 points for family firms versus 0.20 for other firms. When states introduce e-filing for VAT, the effective rate of family firms rises by 0.82 percentage points relative to other firms, closing half the gap.

JEL Classification: H26, G32, O17

Keywords: Family firms, Tax evasion, Ownership concentration, State capacity, India

Megha Patnaik - meghapatnaik@gmail.com
LUISS Guido Carli and CEPR

Acknowledgements
I gratefully acknowledge financial support from the Stanford Institute for Innovation in Developing Economies (SEED) and the International Growth Centre (IGC project 1-VCS-VINC-VXXXX-35305)

Tax, Lies and Red Tape*

Megha Patnaik
LUISS University & CEPR

October 30, 2026

Abstract

Do family firms evade tax when enforcement is weak? I combine a measure of ultimate ownership in India with firm-level excise tax payments and statutory rates. Firms with greater family shareholding have lower effective tax rates. To identify differences in compliance, I use year-by-year changes in statutory rates across products. A one-percentage-point increase in the statutory rate for a firm raises the effective rate by 0.14 points for family firms versus 0.20 for other firms. When states introduce e-filing for VAT, the effective rate of family firms rises by 0.82 percentage points relative to other firms, closing half the gap.

JEL codes: H26, G32, O17.
Keywords: tax evasion, family firms, ownership, state capacity, India.

*LUISS University & CEPR. Email: mpatnaik@luiss.it. I thank Nick Bloom, Petra Moser, and Shai Bernstein for their guidance, and faculty, students and seminar participants at Stanford University for comments and suggestions. I thank the Ministry of Finance, Government of India, and Dr. G.C. Manna for valuable discussions, and participants at the Delhi Macroeconomics Workshop, ISI Delhi, for comments and suggestions. I thank Yashaswini Dunga, Aishwarya Kamboj and Roshni Tara for excellent research assistance. I gratefully acknowledge financial support from the Stanford Institute for Innovation in Developing Economies (SEED) and the International Growth Centre (IGC project 1-VCS-VINC-VXXXX-35305).

1 Introduction

Most firms across the world are family-owned (La Porta, Lopez-de Silanes and Shleifer, 1999). Family ownership concentrates decision-making authority and financial incentives in the hands of controlling families (Bertrand and Schoar, 2006). These families have stronger incentives than diversified owners to reduce tax payments, as they keep a larger share of any tax the firm does not pay. Whether family firms can act on these incentives depends on the capacity of the state. Where state capacity is low and tax enforcement weak, firms have greater opportunities to evade (Besley and Persson, 2009; Pomeranz, 2015). Does concentrated family ownership raise tax evasion where enforcement is weak?

Existing evidence on family firms and taxation focuses on settings with strong enforcement. In this case, family ownership is instead associated with higher tax payments. Chen, Chen, Cheng and Shevlin (2010) find that family firms in the United States are less tax-aggressive than other firms, and attribute this to reputational concerns and penalties from detection. Desai, Dyck and Zingales (2007) show using cross-country data that a higher corporate tax rate raises less revenue where corporate governance is weak. Previous work on developing countries focuses on policies to mitigate evasion, a well-recognised problem in these settings. Pomeranz (2015) shows in Chile that the paper trail of the VAT deters evasion, Naritomi (2019) shows in Brazil that rewarding consumers for demanding receipts raised firms' reported sales, and Basri, Felix, Hanna and Olken (2021) show in Indonesia that moving firms to specialized tax offices doubled the tax they paid. Whether family ownership shapes tax compliance where enforcement is weak, and whether the response to stronger enforcement depends on ownership remains an open question.

In this paper, I study the role of family ownership for tax compliance in India. Family ownership is pervasive among Indian firms (Jameson, Prevost and Puthenpurackal, 2014), which are often organized into family-controlled business groups and have pyramidal ownership structures (Bertrand, Mehta and Mullainathan, 2002). Tax evasion by Indian firms is already documented: Choudhary and Gupta (2025) show that firms slow their reported growth to stay below the excise registration threshold, and Agarwal, Alok, Dixit and Velayudhan (2024) find that firms whose sales become third-party reported stop under-reporting revenue and inflate their wage bills instead. The capacity of the state to detect evasion is limited: the excise intelligence directorate detected Rs. 1,839–2,135 crore (roughly US$0.4–0.5 billion) of excise evasion per year over 2007–2010 through misclassification, undervaluation and the misuse of input-tax credits (Directorate General of Central Excise Intelligence, 2010). In this paper, I begin by examining whether family firms pay lower effective excise rates than

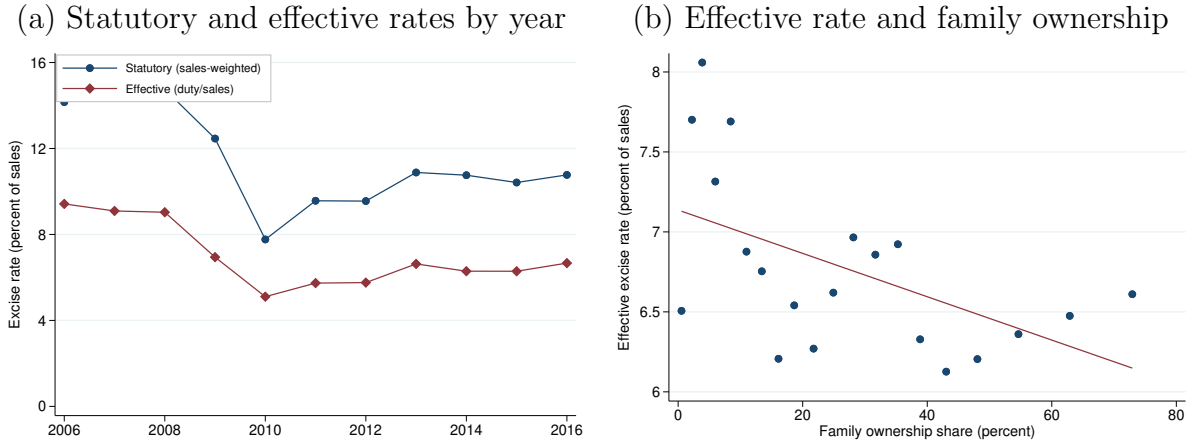
other firms and I find that they do. To identify whether this gap reflects compliance, I then exploit two sources of policy variation. The first is the annual change in statutory excise rates at the product level, announced in the Union Budget each year from 2006 to 2016, which changes the amount of tax firms owe year to year. The second is the staggered introduction of compulsory electronic filing of VAT returns across states between fiscal 2009 and 2016, which changes the likelihood that the authorities detect tax evasion. This is the first firm-level evidence on how ownership shapes compliance where enforcement is weak.

To this end, I construct three new datasets. The first is a firm-year measure of ultimate family control. I identify family ownership through surname and director matching in the shareholding records of the Prowess database published by the Centre for Monitoring Indian Economy (CMIE) and trace ownership through the pyramidal structures of Indian business groups to construct ultimate voting rights under the weakest-link principle. The second is a hand-assembled panel of statutory excise rates by tariff entry and year. This covers over ten thousand entries in over 80 chapters from 2006 to 2016 from the Central Excise Tariff Act schedules and the notifications issued under each Budget. The firm-level statutory rate in a year is then constructed as the average of the rates on the firm's products in that year, weighted by each product's share of its sales. I compare this rate with the firm's effective excise rate, measured as excise duty paid over sales from its audited accounts. The rate panel runs from 2006 to 2016. 2006 is the first full year on the eight-digit tariff, which allows a rate to be assigned consistently to each product.[1] I end the sample in 2016, which is the last year before the Goods and Services Tax replaced the excise duty structure in India. The third dataset I construct records the date on which each of India's states made electronic filing of VAT returns compulsory. This is assembled from state gazette notifications, circulars or published rules.

I preview the findings of the paper in Figure 1. Figure 1(a) shows that the effective rate tracks the statutory schedule over time, with a correlation of 0.97. The statutory rate averages 11.4 percent over 2006 to 2016 and the effective rate 6.8 percent, so all firms on average remit only about three fifths of what the schedule implies. The gap was widest in the years before the stimulus cuts of 2009, at over five points, and narrowest in 2010, at 2.7 points. Figure 1(b) plots a binscatter of firms' mean effective excise rates against their average family ownership across years for the 2,012 manufacturing firms in the cross-section.

[1]The Central Excise Tariff (Amendment) Act, 2004 moved the excise tariff, previously classified at the six-digit level, to the eight-digit code based on the Harmonized System that the customs tariff had used since 2003, so that customs, excise and the trade statistics share one commodity classification, via Notification 6/2005-CE (NT) of 24 February 2005 and applied by field offices under CBEC Circular 808/05/2005-CX of 25 February 2005.

Figure 1: Effective excise rates and family ownership



Notes: Panel (a) plots aggregate statutory and effective excise rates by year over the estimation sample (fiscal 2006–2016), where the statutory rate is the sales-weighted rate implied by each firm's product portfolio. Panel (b) plots mean effective excise rates (excise duty paid over gross sales, winsorized at the 1st and 99th percentiles) against mean ultimate family ownership shares in twenty equal-count bins, with the OLS fit estimated on the underlying 2,012 manufacturing firms (2001–2016). Firm accounts, ownership records, and product sales are from the CMIE Prowess database; statutory rates are from the Central Excise Tariff Act schedules.

The relationship is negative. The effective tax rate for firms falls by 0.14 percentage points for every ten additional points of family ownership, significant at the 1 percent level.

The slope in Figure 1(b) may reflect differences in firm characteristics rather than family ownership. Firms with greater family control are smaller and younger, and may be in industries which differ in their statutory rates and exemptions. I therefore estimate the same cross-sectional regression including firm size, age, listing status and foreign ownership along with industry fixed effects. The coefficient remains stable across specifications including controls. With the full set of controls, a ten percentage point increase in family ownership is associated with a 0.14 percentage point lower effective excise rate, significant at the 1% level. In magnitudes, this is about two percent of the 6.8 percent mean effective excise rate.

The main identification strategy exploits annual changes in product-level excise tax rates announced in the Union Budget. These rates are set at the product level and apply uniformly to all firms producing a specific good. Thus, any individual firm cannot influence the tax rate of its products. The timing and magnitude of the changes are determined by the government's fiscal and industrial-policy objectives rather than by firm-specific conditions. I use this externally determined variation in statutory rates to compare the tax-compliance responses of firms with different degrees of family control. Taking firm product compositions into account, I test whether firms that have greater ultimate family voting rights adjust their

effective tax payments differently from comparable firms when statutory tax rates change. I construct a shift-share measure that holds each firm's product mix fixed before the tax changes, so that exposure cannot respond endogenously to the policy.

The findings are as follows. All firms on average pay less than the statutory rate: a one percentage point increase in the statutory rate raises the effective rate by 0.17 percentage points. Family-controlled firms pay even less. I define blocking family control as a family stake of at least 26 percent, the threshold at which a shareholder can veto a special resolution under the Companies Act and the control threshold of the SEBI takeover code. For firms under blocking control, the effective rate rises by 0.060 less per point of statutory increase (0.14 points against 0.20 for other firms). Exemptions and classifications are set by product and apply to every producer of a good, so the differential arises within the same statutory environment.

My second identification strategy uses the staggered dates at which Indian states made electronic filing of VAT returns compulsory, a measure that increased monitoring by making each firm's reported turnover machine-readable, so that the tax authority can check it against the firm's earlier returns and match the input tax credit it claims against the sales its suppliers report. If family firms remit less because under-reporting goes undetected, their effective rate should rise relative to other firms in the same state once returns become easier to monitor and match. The mandates were issued by the commercial tax departments of the states as their e-filing systems came into force, and they applied to every registered dealer in the state. I compare family-controlled versus other firms facing the same mandate in the same state.

In the states that made electronic filing compulsory, the effective rate of family firms rises by 0.82 percentage points relative to other firms in the same state after the mandate, at a given statutory rate, while the rate of other firms is unchanged. Before the mandate, family firms in these states paid 6.1 percent against 7.8 for other firms, which means the response closes about half of the gap. The gap begins in the year the mandate is introduced and widens over the next three years. The estimate is significant at the 1 percent level on state-clustered standard errors, and at the 10 percent level under randomization inference over the mandate dates.

This paper contributes to three strands of literature. The first is the literature on ownership and corporate governance. Bennedsen, Nielsen, Pérez-González and Wolfenzon (2007) show that passing the chief executive position to a family member lowers firm profitability in Denmark, and Pérez-González (2006) finds the same for inherited control in the United

States. Bertrand et al. (2002) show for India that controlling families move resources toward the group firms in which they have larger cash-flow rights. On taxation, Desai et al. (2007) show that corporate governance changes how tax revenue responds to tax rates, and Chen et al. (2010) find that family firms in the United States are less tax-aggressive than other firms. The twenty-one studies reviewed by Khelil and Khlif (2023) find family firms less tax-aggressive in the United States, Finland and Belgium and more aggressive in India, Brazil and Italy. I contribute to this literature by measuring family control in firms in India in the spirit of Bertrand et al. (2002) and providing the first firm-level evidence that family control predicts that firms pay lower taxes when enforcement is weak.

Second, I contribute to the literature on corporate tax evasion. Allingham and Sandmo (1972) model evasion as a choice that trades off tax saved by the firm against the expected cost of detection, and Crocker and Slemrod (2005) show that corporate evasion also depends on the incentives of the managers who are making tax decisions on behalf of shareholders. Concentrated family ownership closes the owner–manager wedge (Burkart, Panunzi and Shleifer, 2003). My contribution is to study the relation between concentrated ownership and tax evasion in a setting with weak enforcement.

Third, I contribute to the literature on tax enforcement in developing countries. Kleven, Knudsen, Kreiner, Pedersen and Saez (2011) show that evasion is close to zero on income the tax authority observes from third parties and substantial on self-reported income. Pomeranz (2015), Naritomi (2019) and Basri et al. (2021) show that a paper trail, consumer incentives and better-staffed tax offices raise what firms pay. Okunogbe and Pouliquen (2022) find that electronic filing in Tajikistan doubled the tax paid by firms likely to evade, and Kotsogiannis, Salvadori, Karangwa and Murasi (2025) find that e-invoicing in Rwanda raised VAT payments when combined with audits. For India, Choudhary and Gupta (2025) show that firms slow their reported growth to stay below an excise threshold, and Gupta, Velayudhan, Vohra and Garg (2026) find that automated cross-checks of input tax credits cut evasion by 5 percent. My contribution to this set of papers is to show that the incentive matters more where enforcement is weak.

The paper proceeds as follows. Section 2 describes the institutional setting. Section 3 describes the data and the construction of key variables. Section 4 presents cross-sectional evidence. Section 5 presents evidence from statutory rate changes, while Section 6 presents the evidence from e-filing mandates across states. Section 7 concludes.

2 Institutional setting

The central excise tax I study was levied at the product-year level under the Central Excise Act of 1944. It became a value added tax in 1986 when credit for tax paid on inputs was introduced. A national goods and services tax under the Central Goods and Services Tax Act of 2017 replaced it in July 2017. I start the sample in 2006, the first full year after rates were harmonized to the eight digit level to align with the customs tariff, and end it in 2016, the last full year before the transition to the GST began. For each product p that firm i sells in year t, the tax due is the statutory rate applied to the product's sales,

$$\text{Excise tax}_{ipt} = \tau_{pt} \cdot \text{Sales}_{ipt}.$$

This rate τ_{pt} is set centrally in the annual Finance Act enacted with the Union Budget, with specific excise notifications issued under it by the Central Board of Excise and Customs (CBEC), as an ad-valorem rate for each chapter and heading of the Central Excise Tariff Act (CETA).[2] Collection is by self-assessment. The firm classifies its own products in the tariff, computes the duty on each clearance, pays it monthly, and files a monthly return with the Central Excise administration, subject to audit.[3] Education cesses of 2 percent of the duty from 2004 and a further 1 percent from 2007 apply on top of the basic rate, so a 16 percent rate was 16.32 and then 16.48 percent as paid. The cesses were superseded by a higher basic rate in 2016. Three features of this setting are important for the empirical strategy. First, the rate an individual firm faces is set at the product level, and individual manufacturers are small relative to the set of firms that produce a given product. Second, rate changes are hard to anticipate. They are announced in the annual Budget at the end of February or in notifications under it and take effect the same day, so a firm learns its new rate when every other firm does, on the date the Budget is tabled in Parliament. Third, the changes in tax rates originated from broader economic policies rather than from conditions in any one industry.

The largest changes in tax rates during the sample period were responses to the global financial crisis. When it reached India in 2008, the government responded with fiscal stimulus packages in December 2008 and in February 2009 that cut the general excise rate from 14

[2]Products within a chapter typically have the same rate except where the Budget announces an exception. For example, in 2013, chapter 84 on machinery had all product rates set at 12 percent except hand-operated sewing machines, which paid 6 percent. In the same year, chapter 87 on vehicles had a 12 percent rate but motor car rates were set at 24 to 30 percent. This distinction is not crucial for my empirical design.

[3]Manufacturers with clearances below Rs 1.5 crore (Rs 15 million) a year are exempt from duty. Median sales in the sample are Rs 77 crore, about fifty times this limit, and 1.4 percent of firm-years fall below it.

first to 10 and then to 8 percent. The Budgets of 2010 and 2012 then restored the rate to 10 and then 12 percent.[4] Alongside these economy-wide moves, the Budget changed individual chapters for industrial-policy reasons, for example a cut in the rate on pharmaceuticals to 8 percent in 2008 when the general rate stood at 14 percent. In 2008 to 2013 the general rate was therefore 16, 14, 8, 10, 10 and 12 percent, and the rates of most chapters were in line with this.

Operationally, manufacturers had to file periodic (monthly or quarterly) VAT returns with the state's commercial tax department. These returns reported the firm's turnover, the tax due on its sales, and the credit it claimed for tax paid on its purchases. The same turnover amount would appear in the firm's monthly central excise return. Until the late 2000s firms were filing on paper. The introduction of electronic filing meant the return was collected by the state tax department's web portal in a fixed digital format. The department could then machine-read every tax return and receipt and match transactions across buyers and sellers. States introduced electronic filing in different years. Each state first offered e-filing as an option, then required firms above a turnover cutoff to file online, and finally this was applied to all firms. Appendix Table A1 lists the state-wise mandate details.

3 Data

I use four datasets. Two of them come from Prowess, a database of Indian companies compiled by the Centre for Monitoring Indian Economy (CMIE). Prowess accounts for 60 to 70 percent of activity in the organized industrial sector (Goldberg, Khandelwal, Pavcnik and Topalova, 2010). The first dataset is information on shareholding patterns from Prowess. I extract the identities and holdings of major promoters at a quarterly frequency, including corporate owners, which I use to determine firms' ultimate owners. The second is the annual financial statements and product-wise sales of manufacturing firms from Prowess. From this, I extract each firm's excise duty paid along with its product-wise and total sales. I use this to construct the effective excise tax rate. The third is a panel of statutory excise rates by product and year that I construct from annual excise notifications. I match this to each firm's reported products at the product-year level, to compare the statutory rate to the effective rate for observations at the firm-year level. The fourth dataset is a list of the years in which each state made electronic filing of VAT returns compulsory, which I construct from notifications, circulars or rules published by the state tax department.

[4]Explanatory Notes (Central Excise) to the Union Budgets of 2008–09 to 2012–13; Notification 2/2008-CE of 1 March 2008 for the 2008 cut and the pharmaceutical rate; the stimulus notifications of 7 December 2008 and February 2009.

3.1 Family ownership

Founding families own firms both directly and indirectly, often through layers of holding companies (La Porta et al., 1999). Indian business groups are a canonical example: the family sits at the top of a pyramid of holding companies and controls firms in which its direct stake is small (Bertrand et al., 2002; Gopalan, Nanda and Seru, 2007). In my sample, 72 percent of firms have a corporate controlling shareholder. A measure based only on the firm's own shareholders, even after matching shareholders to family names, would therefore understate the family's ultimate stake.

I construct ultimate family ownership by tracing each ownership chain to the individuals at its top and attributing ownership using the weakest-link principle that is standard in the ultimate-ownership literature (La Porta et al., 1999; Claessens, Djankov and Lang, 2000; Faccio and Lang, 2002; Edwards and Weichenrieder, 2004; Di Porto, Pagano, Pezone, Saggio and Schivardi, 2024). This produces a continuous measure of the family's ultimate stake that incorporates both direct and indirect holdings.

This is the first weakest-link measure of ultimate family ownership for India and the first to cover unlisted firms. While previous studies trace ultimate ownership for listed firms or measure the direct stakes of insiders, I trace ultimate stakes in both listed as well as unlisted firms. Masulis, Pham and Zein (2011) trace ultimate ownership for 28,635 listed firms in 45 countries. For India, Jameson et al. (2014) measure founder ownership as the combined stake of the promoters that firms listed on the Bombay Stock Exchange disclose. Kavya and Shijin (2017) identify the ultimate owners of 200 firms in the S&P CNX 500 index in 2015 with the 10 and 20 percent control thresholds of La Porta et al. (1999). Gopalan et al. (2007) cover private as well as public firms in Prowess and measure insider ownership by the combined shareholding of each firm's directors.

I start with data on the percentage holding of the firm's controlling shareholders, the group that the Securities and Exchange Board of India (SEBI) requires firms to identify. I use the latest March (fiscal year-end) snapshot of each firm in each year. Using 4.3 million promoter shareholding records, I classify 20,781 firm-year ownership structures over 2001 to 2016 as follows. For each firm, I first trace the holdings to any ultimate individual owners. When a shareholder is itself a corporation, I match it by name to its own ownership record using the company name, since Prowess does not assign corporate shareholders a company identifier that would allow ready matching back into the Prowess records. The match links the 8,732 companies to their own ownership records using exact name (4,093 firms), exact core name excluding legal suffixes such as Pvt, Ltd and Co stripped (3,476 firms), and blocked fuzzy

matching with character-bigram similarity of at least 0.88 (1,163 firms). For each resolved holding, the tracer opens the holding company's own shareholder list and continues up the pyramid until no chain extends further, ruling out circular ownership. Corporate holdings that cannot be matched to a company with ownership records leave their chains unresolved.[5]

Where a shareholder name identifies the government, a ministry, a bank, an insurer, a mutual or pension fund, a foreign institutional investor, or a venture-capital or private-equity fund, the stake is never attributed to a family and the chain terminates there. Each ultimate owner receives a control share equal to the minimum ownership link along the chain connecting it to the firm, the weakest-link principle of the ultimate-ownership literature (La Porta et al., 1999; Claessens et al., 2000; Faccio and Lang, 2002; Edwards and Weichenrieder, 2004). If individual A owns share s_1 of company B, which owns share s_2 of the firm, then A controls $\min(s_1, s_2)$ of the firm.[6] The procedure follows Di Porto et al. (2024) for Italy and extends it by resolving pyramidal chains in a setting where business-group pyramids are first-order (Bertrand et al., 2002). Holdings of Hindu Undivided Families are attributed to the family. A Hindu Undivided Family is a legal entity under Indian law: lineal descendants of a common ancestor hold ancestral property jointly and are assessed as a single taxpayer. Its beneficial owners are by statute one family, so I treat an HUF's holdings as the family's.

I group ultimate individual owners by family name and define the controlling family as the largest coalition. Grouping by family name requires care in India: initial-based and patronymic naming styles do not place a stable family name in the final position, so the name-parsing step handles honorifics, initials, and HUF suffixes explicitly. Common family names (Jain, Gupta, Patel) mark the merchant communities in which family business is concentrated, so I retain common-name firms only when the controlling name also holds the managing-director or chairperson seat. Women in India usually take their husband's family name at marriage, so daughters who married into other families fall outside the coalition and the traced share understates the family's stake. The family share, FamShare_{it}, is this coalition's weakest-link control share of the firm's total equity.

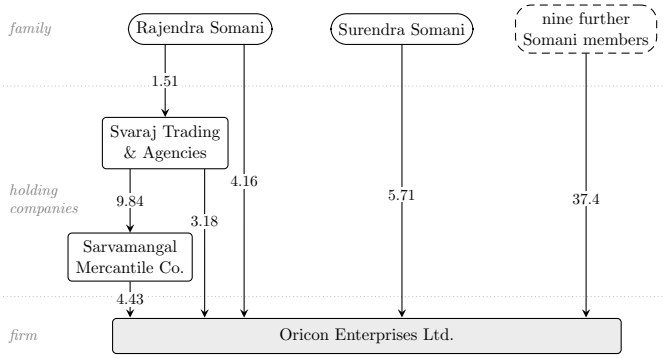
Figure 2 illustrates the procedure with an example from the data. Oricon Enterprises is controlled by the Somani family in 2008 through eleven individual family stakes but also through a pyramid of family holding companies. Stakes held through holding companies

[5]When a control chain ends at a corporate shareholder whose owners I cannot resolve, the block is coded as non-family, the convention of this literature (Di Porto et al., 2024). The traced family share is therefore a lower bound, and misclassification on this margin works against the family differentials I estimate.

[6]The weakest-link share measures control rights. I also compute the family's cash-flow rights as the product of stakes along each chain, summed over the coalition. The two correlate 0.99 because most family control is held directly. I use the control share throughout.

enter with the minimum link along each chain. Rajendra Somani holds 1.51 percent of Svaraj Trading, which holds 3.18 percent of Oricon, so he controls $\min(1.51, 3.18) = 1.51$ percent. The Somani family share of Oricon in 2008 is 63.1 percent. Of this, 47.3 percentage points are held directly and 15.8 through the pyramid structure. Unresolved chains cover only 0.6 percent of equity.

Figure 2: Tracing ultimate ownership: Oricon Enterprises Ltd., 2008



Notes: Selected owners and paths in the ownership structure of Oricon Enterprises Ltd. at fiscal year-end 2008, from Prowess. Arrows point from owners to owned entities and report percentage stakes. The dashed node pools nine further Somani members whose direct stakes sum to 37.4 percent.

For listed firms, the traced family share has a correlation of 0.83 with the individual and HUF equity disclosed to SEBI. Firms the procedure identifies as having higher family ownership are smaller and less productive, in line with the literature (Bertrand and Schoar, 2006; Bloom and Van Reenen, 2007; Di Porto et al., 2024). A ten-percentage-point increase in family control is associated with 15 percent lower sales and 17 percent lower assets, both significant at the one percent level with industry-by-year fixed effects and standard errors clustered by firm (Appendix Table A2).

3.2 Effective rate

The effective excise rate is excise duty paid over sales,

$$\tau_{it}^{\text{eff}} = \frac{\text{Excise duty paid}_{it}}{\text{Sales}_{it}} \times 100, \tag{1}$$

with the numerator and the denominator both taken from the firm's audited annual accounts as reported in Prowess. It is the excise counterpart of the effective tax rate that the corporate tax literature uses to measure tax avoidance (Dyreng, Hanlon and Maydew, 2008; Hanlon and Heitzman, 2010; Chen et al., 2010).

3.3 Statutory rate

I construct a panel of statutory excise rates by tariff entry and fiscal year from 2006 to 2016. The tariff schedule classifies products at the eight-digit level. Products are grouped into 81 chapters covering broad categories of related items such as pharmaceuticals or iron and steel. All manufacturing products, except tobacco and petroleum, which pay specific duties per unit, are included. Rates within a chapter are close to uniform: in the median chapter and year, 98 percent of entries carry the chapter's modal rate. I record the product-year rate using excise notifications from the Central Board of Excise and Customs.

To compute each firm's statutory rate, I match the products for which I have a statutory rate in every year to the products reported by the firms for which I have ownership and accounts data. Firms report their sales by product in their company filings, and Prowess records these product names under its own classification. The sample firms report 3,696 distinct product names in about 623,000 product-year records over 2006 to 2016. The names include "paracetamol", "tractors" and "organic surface-active agents other than soap". They seldom match the wording of tariff entries exactly. I match each product name to the tariff at the product-year level, so every firm that sells the same product in the same year faces the same rate.

The match proceeds in two stages, chapter and then entry within a chapter. The tariff's 10,780 entries sit in 81 chapters, and a product's chapter is far easier to establish than its entry. In the first stage a mapping is created by hand, built independently by two research assistants and reconciled, assigning each product name in the firm product data to a chapter. In the second stage I compare the product name in the firm data with the description of every entry in its chapter in the excise manual. I first drop filler words such as "of", "other" and "parts". Each remaining word gets a weight that rises the fewer entries it appears in, so a distinctive word counts for more than a common one. "Ampicillin" appears in 2 of the 10,780 entries and gets a weight of 8.6, while "acid" appears in about 200 and gets 3.9. An entry's score is the weighted share of the product name's words that the entry's description also contains. A score of 1 means every word of the name appears in the entry. An entry that scores at least 0.45 is a specific match, and the product takes the most common rate among the entries it matches. When no entry reaches 0.45, the product takes the most common rate in its chapter. Rates within a chapter are close to uniform, so this rate is a close substitute. Product names outside the hand mapping receive a chapter from the same score run across all chapters. Pharmaceutical products need one additional step. Firms often name drugs by their use and the tariff names them by their chemical. A short list of synonyms links the two, so that "analgesic" finds the entry for acetylsalicylic acid.

Of the 3,696 names, 1,795 match specific tariff entries, 1,006 take a chapter rate, 764 are services, trade and other activities outside central excise, and 130 are products under specific duties. Weighted by sales value over 2006 to 2016, 24 percent of product sales match a specific tariff entry and 14 percent take a chapter rate, so 38 percent of sales carry an ad valorem rate from the tariff; 52 percent are outside central excise and enter at a zero rate; 9 percent are under specific duties and are excluded from the calculation; the remaining 1 percent are unassigned. A product's rate in a year is the rate on its matched entries in that year, inclusive of the education cess, and a firm's products carry their rates into the firm-year statutory rate of equation (2).

Based on the products the firm produces, a firm's statutory liability in a year is the sum over its products of the product's rate times the product's sales, and its statutory rate is this liability divided by its total sales,

$$\bar{\tau}_{it} = \frac{\sum_p \tau_{pt} \, \text{Sales}_{ipt}}{\sum_p \text{Sales}_{ipt}}, \tag{2}$$

the excise-to-sales ratio a fully compliant firm with firm i's product mix would pay. Excise is self-assessed: the firm classifies its own products in the tariff and computes rate times sales, product by product, which is what the tariff schedules exist for. The statutory rate is therefore the liability implied by the firm's reported product mix and the tariff schedule. Product mix matters for it: 91 percent of firm-years in the design sample produce more than one product, with a median of 5 products and a maximum of 86. Prowess dates each firm-year by its accounting close, and I keep firms closing in March, so that firm-year t runs from April of $t-1$ to March of t. A rate set in the Budget of February $t-1$ governs fiscal year t, and a mid-year change is assigned to the fiscal year it governed for most months. The estimation sample keeps the firms whose statutory rate is reliably measured: products in the hand-coded map make up at least 60 percent of the firm's dutiable product sales, weak automatic matches at most 10 percent, and dutiable products at least half of its product sales. This rule keeps 10,654 of the 35,618 firms with product data.

As an example, Table 1 shows rates for four products through the sample period: the 2008–09 stimulus cuts and the rollbacks in the standard-rate products, the 2008 pharmaceutical cut to 8 percent one year ahead of the general trough, and the concessional track for biscuits.

The rate-matched panel is the 27,080 firm-years from 5,013 firms with a reliably matched statutory rate over 2006 to 2016. In this sample, the effective excise rate has mean 6.8 percent, median of 7.2 and interquartile range from 3.0 to 10.1. Most firms (family and non-family) pay well below the standard statutory rate. The statutory rate implied by their

Table 1: Statutory excise rates by year, selected products

Product	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Soap	16.32	16.48	16.48	14.42	8.24	10.30	10.30	12.36	12.36	12.36	12.50
Tractors	16.32	16.48	16.48	14.42	8.24	10.30	10.30	12.36	12.36	12.36	12.50
Paracetamol	16.32	16.48	16.48	8.24	8.24	10.30	10.30	6.18	6.18	6.18	6.00
Biscuits	16.32	8.24	8.24	8.24	4.12	4.12	5.15	6.18	6.18	6.18	6.00

Notes: Statutory excise rate in percent, inclusive of education cesses, by year ending March, for four Prowess product names matched to specific tariff entries (chapters 34, 87, 30 and 19). Source: CBEC excise notifications.

products averages 11.4 percent, so the effective rate is on average 4.6 percentage points below the statutory rate.

3.4 Electronic filing mandates

The second design needs the year in which each state made electronic filing of VAT returns compulsory for all firms. Each state set this date in its own legal instrument, a notification, a circular or an amendment to its VAT rules. I collect these instruments from the state commercial tax departments and from the technical guides to the state VAT laws published by the Institute of Chartered Accountants of India. For many states the instrument itself is hard to find, and secondary sources place the mandate in different years. I therefore use a state's date only when I have read it in the instrument. Appendix Table A1 lists every state with its instrument and date.

This rule sorts the states into three groups. Nine states have a documented mandate within the sample period. Six of them form the main sample: Maharashtra, Kerala, Rajasthan, Andhra Pradesh, Odisha and Jharkhand. In each of the six, a single instrument covers all firms and the sample has at least two years after the mandate. The extended sample adds the other three, Delhi, Gujarat and Haryana. In Delhi the mandate reaches all firms only through two instruments together, an order of 2007 and a rule of 2013. Gujarat and Haryana mandated in 2016, the last year of the sample. Thirteen other states also mandated electronic filing within the sample period, but their dates come only from departmental web pages, practitioners' accounts or secondary sources.

3.5 Summary statistics

Table 2 reports summary statistics for the design sample. The effective excise rate averages 6.3 percent of sales, against a contemporaneous statutory rate of 11.5 percent and a predetermined exposure of 11.5. The median firm has sales of Rs 1.7 billion and total assets of Rs

Table 2: Summary statistics, design sample

	Mean	SD	p10	Median	p90	N
Effective excise rate (% of sales)	6.31	4.18	0.45	6.56	11.80	6,326
Statutory rate, contemporaneous (%)	11.50	3.42	7.36	11.74	16.32	6,317
Exposure, predetermined (%)	11.48	3.41	7.27	11.73	16.29	6,326
Family control share, 2005–07 (%)	21.21	18.44	1.40	16.04	46.86	6,326
Family cash-flow share, 2005–07 (%)	20.62	18.50	1.04	15.33	46.18	6,326
Blocking control (share $\geq$ 26%)	0.35	0.48	0.00	0.00	1.00	6,326
Sales (Rs mn)	8,528	29,660	198	1,742	17,537	6,326
Total assets (Rs mn)	11,597	54,066	187	1,609	20,263	6,326
Firm age (years)	33.10	17.65	17.00	28.00	58.00	6,326
Listed	0.82	0.38	0.00	1.00	1.00	6,326

Notes: Design sample, 6,326 firm-years from 709 firms, 2006 to 2016. The effective excise rate is excise duty over gross sales, winsorized at the 1st and 99th percentiles. The contemporaneous statutory rate and predetermined exposure are defined in Section 5; the contemporaneous rate is missing for 9 firm-years. Family control and cash-flow shares are firm means over 2005 to 2007; blocking control is a share of at least 26 percent. Sales and assets in millions of rupees.

1.6 billion, and is 28 years old. Family control varies widely across firms. The family control share averages 21 percent with a median of 16. A tenth of firm-years have a share below 1.4 percent and a tenth have a share above 47 percent. The cash-flow share is within a point of the control share throughout the distribution. Blocking control, a share of at least 26 percent, covers 35 percent of firm-years and 252 of the 709 firms. Majority control, a share above 50 percent, covers 8 percent of firm-years and 58 firms.

4 Effective rates across firms

4.1 Specification

First, I examine how effective tax rates vary with family control. Existing research on ownership and taxation is based on cross-sectional regressions of a firm's effective tax rate on a measure of its ownership. In the United States, Chen et al. (2010) find that family firms pay higher effective tax rates than non-family firms, and Badertscher, Katz and Rego (2013) find that firms with concentrated ownership and control avoid less tax. Both results come from a setting where enforcement is strong and the margin of adjustment is lawful avoidance.

I follow the literature and estimate the following cross-sectional OLS specification:

$$\tau_i^{\text{eff}} = \alpha + \beta \, \text{FamShare}_i + \delta' X_i + \eta_{j(i)} + \varepsilon_i, \tag{3}$$

where τ_i^{eff} is the firm's mean effective excise rate over the sample years, FamShare_i is the ultimate family control share of Section 3, X_i includes log assets, firm age, a listed indicator and a foreign-ownership indicator, and $\eta_{j(i)}$ is a fixed effect for the firm's two-digit industry. I include industry fixed effects, thus comparing firms in the same two-digit division of the National Industrial Classification (NIC), which are likely to make similar products and thus face similar statutory rates. I include controls for size and age. Standard errors are robust to heteroskedasticity.

4.2 Results

Table 3 reports estimates of equation (3). Column (1) replicates the slope in Figure 1(b). The coefficient on family control is -0.014, significant at the 1 percent level. A ten-point increase in family control is associated with an effective excise rate 0.14 percentage points lower, about 2 percent of the sample mean of 6.8 percent. Column (2) adds industry fixed effects, and the coefficient remains stable at -0.014, still significant at the 1 percent level. Column (3) adds firm size, measured as log assets, and the coefficient is -0.016. Column (4) adds firm age, and the coefficient is -0.015. Column (5) adds an indicator for listed firms, and the coefficient is -0.013. Column (6) adds an indicator for foreign ownership, and the coefficient is -0.014. Across the columns the coefficient moves slightly, but remains stable in magnitude and significant at the 1 percent level.

5 Responses to statutory rate changes

A lower effective rate can arise from evasion, from lawful reductions in duty through exemptions, input credits or product classification, or even from differences across firms that the controls cannot fully account for. To causally identify the relationship between family control and evasion, I use changes in statutory excise rates and examine how firm responses vary by ownership.

5.1 Specification

A firm's exposure is its pre-reform product mix at each year's statutory rates. Let $s_{ip,\,05\text{–}07}$ be the share of product p in firm i's sales averaged over 2005 to 2007, and τ_{pt} the national statutory rate on product p in year t. Exposure is measured as

$$\text{Exposure}_{it} = \sum_{p} s_{ip,\,05\text{–}07}\ \tau_{pt}. \tag{4}$$

Table 3: Family control and the effective excise rate

	Effective excise rate (percent of sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
Family ownership share	-0.014***	-0.014***	-0.016***	-0.015***	-0.013***	-0.014***
	(0.005)	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)
Log assets			-0.111**	-0.150***	-0.142**	-0.154***
			(0.055)	(0.057)	(0.057)	(0.056)
Firm age				0.016***	0.016***	0.015***
				(0.006)	(0.006)	(0.006)
Listed					-0.272	-0.206
					(0.218)	(0.220)
Foreign/MNC						1.091**
						(0.432)
Industry FE	No	NIC-2	NIC-2	NIC-2	NIC-2	NIC-2
Number of firms	2,012	2,012	2,012	2,012	2,012	2,012
Firm-year observations	20,177	20,177	20,177	20,177	20,177	20,177

Notes: Firm-level cross-section of 2,012 manufacturing firms, 2001–2016, each observed as its mean over the 20,177 firm-years in the panel. The dependent variable is the firm's mean effective excise rate, excise duty over sales in percent, winsorized at the 1st and 99th percentiles. Estimates of equation (3). Column (1) is the bivariate regression; Columns (2) to (6) add NIC two-digit industry fixed effects, log assets, firm age, a listed indicator and a foreign-owned indicator, one at a time. Family control share is the ultimate family voting share of Section 3, in percentage points. Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

I estimate

$$e_{it} = \alpha_i + \lambda_t + \beta \, \text{Exposure}_{it} + \delta \left(\text{Family}_i^{\geq 26} \times \text{Exposure}_{it} \right) + \varepsilon_{it}, \tag{5}$$

where e_{it} is the effective excise rate of firm i in year t. The coefficient β is passthrough, the share of a one-point change in the statutory rate that reaches the rate the firm pays. Under full compliance, every firm remits the duty it owes and β equals one. The coefficient δ is the family differential: any additional passthrough (positive or negative) of firms under family blocking control. Regressions include firm fixed effects, which absorb each firm's average gap between its effective and statutory rate, and year fixed effects, which absorb the national path of rates, including the stimulus cuts of 2008 and 2009 and their withdrawal. The estimates are identified from differences in the size and timing of rate changes across products. Standard errors are clustered on the firm's pre-reform dominant tariff chapter, the level at which rates are broadly set, which gives 60 clusters.

I take a cutoff for family ownership, $\text{Family}_i^{\geq 26}$, corresponding to at least 26 percent family control share, as this identifies blocking control or veto power. A special resolution needs

Table 4: Statutory rate changes and the family differential

	Predetermined exposure		Contemporaneous statutory rate	
	(1)	(2)	(3)	(4)
Statutory rate measure	0.173***	0.197***	0.173***	0.191***
	(0.030)	(0.030)	(0.026)	(0.026)
× Family ≥ 26%		-0.060**		-0.048**
		(0.023)		(0.023)
Firm-year observations	6,302	6,302	6,291	6,291

Notes: Sample consists of 6,302 firm-years from 685 firms from years 2006 to 2016. The dependent variable is the effective excise rate, excise duty over gross sales, winsorized at the 1 percent level. Exposure is the firm's 2005 to 2007 product mix at each year's statutory rates. Columns (3) and (4) use the contemporaneous statutory rate. All columns include firm and year fixed effects. Standard errors in parentheses are clustered on the firm's pre-reform dominant tariff chapter (60 clusters). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

75 percent of the votes cast, so a stake above 25 percent can block it.[7] I measure the share as fixed over 2005 to 2007, the same years I use for the exposure variable. The share is persistent within firms. Its 2005 to 2007 value has a correlation of 0.84 with the share in each year.

5.2 Results

Table 4 reports estimates of equation (5). In Column (1), I begin by regressing the effective excise rate on exposure, with firm and year fixed effects, and find a coefficient of 0.173, significant at the 1 percent level. This implies that a one-point increase in predetermined exposure raises the effective excise rate by 0.173 percentage points. In Column (2), I add the interaction of exposure with family blocking control. Passthrough for firms below the 26 percent threshold is 0.197, and firms under family blocking control pass through 0.060 less, significant at the 5 percent level, giving them a passthrough of 0.137. The differential is about a third of average passthrough. Following a five-point increase in the statutory rate, the effective rate of a family firm rises by about 0.3 percentage points less than that of a comparable non-family firm.

The identifying assumption for δ to be causal is that when the statutory rate on a good rises, the duty owed per rupee of sales rises by the same amount for family firms and for other firms. The assumption fails if family firms and other producers of the same good have

[7] Companies Act 2013, section 114(2). This is a standard control cutoff, for example also used in takeover regulations. Acquiring 25 percent of the voting rights in a listed company obliges the buyer to make an open offer to the other shareholders (SEBI (Substantial Acquisition of Shares and Takeovers) Regulations 2011, regulation 3).

different lawful ways to respond. The main such margin is switching sales toward products whose rate has fallen or stayed low. For this reason, I re-estimate the equation on the contemporaneous product mix of the firm. A second margin is exports, which are exempt from excise. A firm that exports more passes less of a rate change into its effective rate, so Appendix Table A3 re-estimates the equation on low exporters. Under this assumption the family differential is a difference in remittance. When the statutory rate rises, family firms remit less of the increase than the law prescribes.

Columns (3) and (4) repeat the two specifications with the contemporaneous statutory rate,

$$\tau_{it}^{\text{con}} = \sum_{p} s_{ipt}\ \tau_{pt}, \tag{6}$$

built from the shares s_{ipt} of the products the firm actually makes in each year, the firm-year statutory rate as before. Passthrough is 0.173, the same as on predetermined exposure, significant at the 1 percent level. Firms under family blocking control pass through 0.048 less, significant at the 5 percent level, against 0.060 on predetermined exposure. The family differential is similar in size whether the rate is measured on the predetermined product mix or on the contemporaneous one. This is to be expected, since if firms change products, they typically change them within chapter: the average firm earns 95 percent of its sales in a single tariff chapter.

I run several robustness tests, reported in Appendix Table A3. First, gross sales include the excise duty itself, so I recompute the effective rate over sales net of duty. With this, the differential increases to -0.079. Next, I examine how the results change with different samples. Among listed firms, whose accounts face the closest scrutiny and where evasion would be much harder, the differential is -0.056 and still significant at the 5 percent level. Next, I exclude the stimulus years of 2009 to 2011, so that the estimate is determined instead by smaller rate changes. In this case, the coefficient on the interaction of family control with exposure is -0.070. I also keep only firms whose pre-reform sales fall in a single tariff chapter, and find a coefficient of -0.059 with a standard error of 0.040. Next, I exclude textiles, where producers could opt out of duty by forgoing input credits, so a low effective rate there can be a lawful choice, and find a coefficient of -0.075, significant at the 1 percent level. I also examine firms that export less than a quarter of their sales, which has a coefficient of -0.081, significant at the 10 percent level. Clustering by chapter and year together leaves the baseline significant at the 5 percent level. The effective rates of family firms could also drift over time. Any drift that tracks the path of rates would look like a smaller response. I check for this in Columns (8) and (9), where I add a linear trend for each firm and, separately, for

each chapter. The differential is −0.064 in both cases.

The 60 chapter clusters are of unequal size, and conventional cluster-robust inference can over-reject in such settings (Cameron, Gelbach and Miller, 2008). For this reason, I also run the regression with a wild cluster bootstrap applied by chapter, with the null imposed. The bootstrap gives $p = 0.015$ against a conventional p of 0.012.[8] I finally also check whether any one chapter is driving the estimate. Dropping each chapter in turn leaves the differential between −0.050 and −0.072, significant at the 5 percent level throughout.

Rates fell and rose during the sample period, and firms' reactions to cuts and hikes may differ. In Appendix Table A4, I estimate the response to cuts and to hikes separately. I take the 285 chapter rate changes of 2 to 12 points, and for each one compare firms in that chapter with firms in chapters whose rate did not change. For each point of a rate change, the effective rate falls by 0.267 for the 140 cuts and rises by 0.151 for the 145 hikes, both significant at the 1 percent level. Firms therefore pass on more of a cut than of a hike. This is consistent with a story of evasion: a firm that conceals part of its sales owes less when the rate falls and the reduction shows up in the effective rate. However, when the rate rises the increase in the effective rate is not proportionate.

6 Responses to electronic filing mandates

Family firms have the incentive to evade, but they can act on it only when enforcement is weak. I examine whether the difference in tax payments between family and non-family firms shrinks when enforcement strengthens, using the introduction of mandatory electronic filing of VAT in different states in different years.

6.1 Specification

Mandates arrive in different years, so I use a stacked design. For each adoption year e I form a dataset of the firms registered in the states first mandated in e and of control firms, those registered in states still unmandated at $e + 4$, over fiscal years $e - 4$ to $e + 4$. Each dataset is one event year. Pooling the event years, I estimate

$$e_{it}^{(e)} = \alpha_i^{(e)} + \lambda_t^{(e)} + \gamma \left(M_s \times \text{Post}_t^{(e)} \right) + \delta \left(\text{Family}_i^{\geq 26} \times M_s \times \text{Post}_t^{(e)} \right) + \varepsilon_{it}^{(e)}, \qquad (7)$$

[8] The bootstrap draws 9,999 samples. Each sample reweights the residuals of every chapter by a random draw from the six-point distribution of Webb (2023), which is designed for a few dozen clusters.

where e_{it} is the effective excise rate of firm i in year t, M_s indicates firms registered in a state mandated in e, $\text{Post}_t^{(e)}$ equals one from year e, and $\alpha_i^{(e)}$ and $\lambda_t^{(e)}$ are firm and year effects specific to each event year. $\text{Family}_i^{\geq 26}$ is defined as the blocking-control indicator, as in the previous section. The coefficient γ is the change in the effective rate of other firms after the mandate. The differential is measured as the coefficient δ, the additional response of family firms. I estimate the differential within a given treatment year and within treated states of that year. A state-level shock that coincides with adoption moves both groups and is therefore absorbed by the mandate's main effect, γ. The identifying assumption is that family and other firms in the same state would move in a similar manner in the absence of a mandate. In a second specification, I also include the firm's contemporaneous statutory rate as a regressor, so that the mandate's effect is measured at a given statutory rate.

To measure effects over time as well as test for pre-trends, I also estimate the event-study form of the above specification,

$$e_{it}^{(e)} = \alpha_i^{(e)} + \lambda_t^{(e)} + \sum_{k \neq -1} \gamma_k \left(M_s \times \mathbb{1}\{t - e = k\} \right) + \sum_{k \neq -1} \delta_k \left(\text{Family}_i^{\geq 26} \times M_s \times \mathbb{1}\{t - e = k\} \right) + \varepsilon_{it}^{(e)}, \tag{8}$$

where k runs from -4 to 4 and the year before the mandate, $k = -1$, is the reference year. The coefficients δ_k for $k < -1$ test for a differential pre-trend between family and other firms, and those for $k \geq 0$ trace the differential after the mandate is introduced.

I estimate the design on two samples. The main sample is the six states whose mandate I read from a single instrument that made electronic filing compulsory for all firms: Maharashtra, Kerala, Rajasthan, Andhra Pradesh, Odisha and Jharkhand. The extended sample adds Gujarat, Haryana and Delhi, where the window of the study differs from that of the main sample. In Gujarat and Haryana the mandate takes effect in 2016, the last year of the sample, so I am limited to only one year after the mandate. In Delhi, a 2007 order made electronic filing compulsory for firms that filed quarterly, followed by a 2013 rule which put every firm on quarterly filing, so the mandate covered all firms only from 2013. Delhi therefore has four years after the mandate, like the main-sample states, but some of its firms had already filed electronically for six years before it. The main sample has 306 firms in six event years and the extended sample 506 firms in eight. Standard errors are clustered by state, the level at which mandates are set.

Table 5: Electronic filing mandates and the family differential

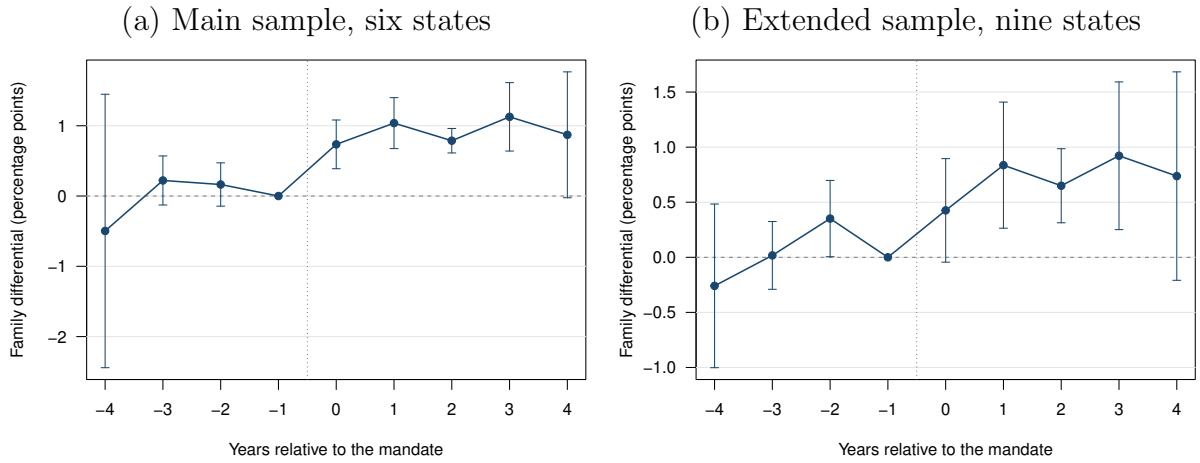
	Main sample (six states)				Extended sample (nine states)			
	All event years		Event years with 30+ treated firms		All event years		Event years with 30+ treated firms	
	(1) Effective rate	(2) Statutory control	(3) Effective rate	(4) Statutory control	(5) Effective rate	(6) Statutory control	(7) Effective rate	(8) Statutory control
Mandate × Post	0.152	0.262	-0.047	0.266	-0.258	-0.227	-0.237	-0.112
	(0.431)	(0.420)	(0.499)	(0.487)	(0.176)	(0.218)	(0.184)	(0.224)
× Family ≥ 26%	0.837***	0.824***	0.895***	0.873***	0.651**	0.589**	0.679**	0.609**
	(0.194)	(0.197)	(0.154)	(0.167)	(0.234)	(0.249)	(0.233)	(0.255)
Statutory rate, contemporaneous		0.171***		0.173***		0.167***		0.160***
		(0.014)		(0.015)		(0.009)		(0.012)
Randomization inference p	0.09	0.08			0.02	0.04		
State clusters	9	9	7	7	12	12	10	10
Stacked firm-year observations	2,564	2,373	2,095	1,905	6,733	6,418	4,172	3,867

Notes: The dependent variable is the effective excise rate, excise duty over gross sales in percent, winsorized at the 1st and 99th percentiles. Family blocking control is a 2005 to 2007 family control share of at least 26 percent. Columns (1) to (4) use the main sample (306 firms, six event years); Columns (5) to (8) use the extended sample, which adds Gujarat, Haryana and Delhi (506 firms, eight event years). Columns (3), (4), (7) and (8) keep only the event years with at least 30 treated firms. Standard errors in parentheses are clustered by state. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.2 Results

Table 5 reports estimates of equation (7). Columns (1) to (4) run the regression on the main sample and Columns (5) to (8) on the extended sample. In Column (1), the effective rate of other firms rises by 0.15 percentage points after the mandate, and the rate of family firms rises by a further 0.84 points, with the differential for family firms significant at the 1 percent level. The effect on other firms is small and insignificant. This is consistent with family firms evading tax more than other firms in the absence of strict enforcement. Electronic filing raises the cost of concealed sales through higher chances of detection, and firms that were already compliant should show no effect. Before the mandate, family firms in these states pay an effective rate of 6.08 percent against 7.79 for other firms. This means that the response closes about half of the gap between the two groups. In Column (2), I also add the contemporaneous statutory rate. The coefficient on the statutory rate is 0.17 (0.01), the same response as in the previous section. The coefficient on the interaction remains 0.82, significant at the 1 percent level. Columns (3) and (4) keep only the event years with at least 30 treated firms. Treated firms are spread unevenly across states. Maharashtra has 213 and Andhra Pradesh 50, while the other four states in the main sample have fewer than 20 each. With only Maharashtra and Andhra Pradesh the coefficient on the interaction is 0.90 and 0.87, significant at the 1 percent level. Columns (5) to (8) repeat the four specifications on the extended sample. The coefficient on the interaction is 0.65 in Column (5) and 0.59 in Column (6), both significant at the 5 percent level. Before the mandate, family firms in

Figure 3: Family differential around electronic filing mandates



Notes: Each panel plots the coefficients δ_k from equation (8) with the contemporaneous statutory rate as a regressor, the year before the mandate as the reference, and 95 percent confidence intervals from standard errors clustered by state. Panel (a) uses the main sample of Table 5, Column (2), and panel (b) the extended sample of Column (6). The three pre-mandate coefficients are jointly insignificant in both panels ($p = 0.19$ in panel (a), $p = 0.21$ in panel (b)).

the nine states pay 6.21 percent versus 7.12 for non-family firms, so the introduction of the mandate closes about two thirds of the gap. Columns (7) and (8) again restrict the sample as in Columns (3) and (4) to event years with at least 30 treated firms. The coefficient on the interaction is 0.68 and 0.61 respectively, significant at the 5 percent level.

With only nine clusters in the main sample, conventional clustering can overstate precision. I therefore also report a randomization inference p-value for the family differential. I calculate this by reassigning the mandate years at random across the states, keeping the set of dates fixed, re-estimating the differential, and repeating this 500 times. The p-value is then the share of these random assignments that produce a differential at least as large as the estimate. In the main sample the randomization inference p-value is 0.09 for the coefficient of 0.84 in Column (1), and 0.08 for the coefficient of 0.82 in Column (2). In the extended sample, the p-value is 0.02 for the coefficient of 0.65 in Column (5) and 0.04 for the coefficient of 0.59 in Column (6). The main-sample differential is therefore significant at the 10 percent level under randomization inference and the extended-sample differential at the 5 percent level.

Figure 3 plots the coefficients of equation (8), and Appendix Table A5 reports the corresponding coefficients. Panel (a) shows the event study for the main sample. In the three years before the mandate the family differential is close to zero, and the three pre-mandate coefficients are jointly insignificant ($p = 0.19$). In the mandate year the differential rises to 0.74 percentage points. Over the next three years it is 1.04, 0.79 and 1.13, each significant at the 1 percent level. The third-year coefficient exceeds the mandate-year coefficient by 0.39,

significant at the 1 percent level, i.e. the differential widens significantly after the mandate, rather than increasing upon introduction and then flattening. Averaged over the mandate year and the four years after it, the differential is 0.91 points, about half of the 1.71-point gap between family and non-family firms before the mandate. In contrast, the effective rate of non-family firms does not move significantly in any year. This is consistent with evasion being higher in family firms. Non-family firms appear to have reported most of what they owed before the mandate, so the e-filing requirement had little effect on their payments. The event study is run on the extended sample in Panel (b) and demonstrates the same pattern. The difference in effective tax rates between family and non-family firms is 0.43 in the mandate year and increases to 0.84, 0.65 and 0.92 over the next three years. The pre-mandate coefficients are -0.26, 0.02 and 0.35, jointly insignificant ($p = 0.21$).

7 Conclusion

In this paper, I ask whether family ownership shapes tax compliance of firms when enforcement is weak. I study India's central excise duty with a new measure of each firm's ultimate family control and a panel of the statutory rates each firm owes on its products. I find that family firms pay less. In the cross-section, a ten percentage point increase in family ownership is associated with an effective rate 0.14 percentage points lower. A one percentage point rise in the statutory rate raises the effective rate by 0.17 points on average, by 0.14 points for family firms and by 0.20 for other firms. With the introduction of e-filing of VAT made mandatory by states, detection is easier, and I find that family firms evade less than before. Their effective rate rises by 0.82 percentage points relative to other firms in the same state, closing about half of the gap.

Two conditions are important for the result of evasion by family firms. First, controlling families have high-powered incentives, since they keep a large share of any costs the firm saves, including duties. The second is that the firms have to operate in an economic environment with weak enforcement. In cases where enforcement is strong, the relationship has been found to go the opposite direction: Chen et al. (2010) find that family firms in the United States engage in less tax avoidance. This is because to reduce costs, family firms engage in lawful avoidance, but this still carries reputational implications which can harm firm value. This difference across countries highlights the role of state capacity in tax compliance, over and above the role of ownership structure.

A Additional Tables

Table A1: Electronic filing mandates by state

State	Instrument making electronic filing compulsory	First year
Maharashtra	Rule 17 amended by notification of 14 March 2008; all dealers from 1 April 2009 (Trade Circular 6T of 2009)	2009
Kerala	Commercial Taxes Department: e-filing for all registered dealers from 1 January 2009	2010
Rajasthan	Notifications of 29 June and 9 December 2010	2011
Andhra Pradesh	Circular 14 of 7 May 2011, all dealers	2012
Odisha	Notification 18837 of 5 September 2013, effective 1 October 2013	2014
Jharkhand	S.O. 57 of 22 October 2014, all registered dealers	2015
Delhi	Commissioner's order of 2007, electronic return for quarterly dealers, read with the Amendment Rules of 30 March 2013, tax period a quarter for all dealers, effective 1 April 2013	2013
Gujarat	Rule 19(8A) inserted by Notification GHN-19/VAR-2014 of 15 October 2014	2016
Haryana	Section 54A (2015) and Commissioner's orders of 5 August 2015	2016
Uttar Pradesh	Department web page: turnover above Rs 1 crore from 2009–10; instrument unlocated	2010
Bihar	Mandate undated	2011
Karnataka	Notifications KSA/CR-60/2010-11 of 2010; dealer coverage unread	2011
Madhya Pradesh	Rule 23-A of 18 May 2010; operative notification unlocated	2011
Dadra and Nagar Haveli	Coded 2011, unverified	2012
Daman and Diu	Coded 2011, unverified	2012
Goa	Coded 2011, unverified	2012
West Bengal	Selected-dealer e-filing from 2010; no all-dealer mandate located	2012
Manipur	Coded 2012, unverified	2013
Punjab	Online returns after 31 October 2012; instrument unlocated	2013
Uttarakhand	Department site, May 2012; instrument unlocated	2013
Assam	Coded 2014, unverified	2015
Himachal Pradesh	Coded 2014, unverified	2015
Tamil Nadu	e-C Tax portal, January 2016; after the sample period	2017
Never mandated: controls in every stack.		
Chandigarh	No mandate on record	
Chhattisgarh	No mandate located	
Puducherry	No mandate on record	

Notes: The first year is the first full year under the mandate. Each firm is assigned to the state of its registered office. Telangana is included into Andhra Pradesh to maintain one state assignment over the sample period.

Table A2: Family ownership and firm performance

	(1) Log sales	(2) Log assets	(3) Log employment	(4) Log sales per employee	(5) Log value added per employee
Family control share	-0.016***	-0.019***	-0.015***	-0.003***	-0.003**
	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)
Industry × year FE	Yes	Yes	Yes	Yes	Yes
Firm-year observations	13,966	13,966	5,320	5,320	5,294

Notes: Sample is the Prowess firm-year panel from 2001 to 2016. Value added is sales minus raw materials, stores and spares, and power, fuel and water. All columns include industry × year fixed effects at the two-digit NIC level. Columns (3) to (5) use the subsample for which employee count is available. Standard errors in parentheses are clustered by firm. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3: Response to statutory rate changes using alternative samples and specifications

	(1) Net of duty	(2) Listed firms	(3) Excluding 2009–11	(4) Single chapter	(5) Excluding textiles	(6) Low exporters	(7) Two-way clustering	(8) Firm trends	(9) Chapter trends
Panel A: Predetermined exposure									
Statutory rate measure	0.241***	0.209***	0.201***	0.179***	0.187***	0.310***	0.197***	0.155***	0.161***
	(0.035)	(0.027)	(0.034)	(0.049)	(0.045)	(0.041)	(0.040)	(0.030)	(0.026)
× Family ≥ 26%	-0.079***	-0.056**	-0.070**	-0.059	-0.075***	-0.081*	-0.060**	-0.064**	-0.064***
	(0.028)	(0.027)	(0.030)	(0.040)	(0.025)	(0.047)	(0.020)	(0.027)	(0.022)
Passthrough, family firms	0.161	0.154	0.131	0.120	0.112	0.229	0.137	0.091	0.096
(standard error)	0.041	0.034	0.046	0.054	0.051	0.055	0.031	0.029	0.028
Firm-year observations	6,302	5,200	4,597	2,960	5,569	2,991	6,302	6,302	6,302
Panel B: Contemporaneous statutory rate									
Statutory rate measure	0.231***	0.199***	0.199***	0.167***	0.186***	0.291***	0.191***	0.165***	0.169***
	(0.030)	(0.026)	(0.029)	(0.043)	(0.036)	(0.037)	(0.028)	(0.030)	(0.026)
× Family ≥ 26%	-0.063**	-0.050*	-0.056**	-0.045	-0.059**	-0.068	-0.048*	-0.055*	-0.050**
	(0.029)	(0.025)	(0.027)	(0.038)	(0.026)	(0.043)	(0.022)	(0.027)	(0.022)
Passthrough, family firms	0.168	0.149	0.143	0.122	0.128	0.223	0.143	0.111	0.119
(standard error)	0.036	0.032	0.039	0.048	0.042	0.044	0.028	0.030	0.028
Firm-year observations	6,291	5,194	4,586	2,951	5,558	2,987	6,291	6,291	6,291

Notes: The dependent variable is the effective excise rate. Panel A uses predetermined exposure while Panel B uses the contemporaneous statutory rate. All columns include firm and year fixed effects. Standard errors in parentheses are clustered on the firm's pre-reform dominant tariff chapter, and in Column (7) on chapter × year. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4: Response to rate cuts and rate hikes

	(1) All changes	(2) Rate cuts	(3) Rate hikes
ΔRate × Post	0.249***	0.267***	0.151***
	(0.033)	(0.039)	(0.056)
Rate-change events	285	140	145
Tariff chapters (clusters)	60	58	59
Stacked firm-year observations	28,063	15,380	12,683

Notes: The dependent variable is the effective excise rate. Each rate change is stacked with a window of three years before and after the change. Control firms are those whose pre-reform dominant chapter kept its rate within one percentage point of its starting value over the window. The regressor is the rate change in percentage points, negative for a cut and positive for a hike, for treated firms in the years after the change. All columns include firm-by-event and year-by-event fixed effects. Standard errors in parentheses are clustered on tariff chapter. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A5: Event-study coefficients around electronic filing mandates

	Main sample		Extended sample	
Relative year	Family differential	Other firms	Family differential	Other firms
$t = -4$	−0.497	−0.318	−0.259	0.087
	(0.993)	(0.552)	(0.379)	(0.302)
$t = -3$	0.221	−0.312	0.017	0.031
	(0.178)	(0.381)	(0.157)	(0.227)
$t = -2$	0.164	−0.202	0.352*	−0.141
	(0.157)	(0.234)	(0.177)	(0.124)
$t = 0$	0.735***	−0.023	0.426	−0.049
	(0.177)	(0.270)	(0.240)	(0.138)
$t = 1$	1.038***	−0.191	0.836**	−0.263
	(0.185)	(0.511)	(0.292)	(0.189)
$t = 2$	0.787***	0.026	0.650***	−0.372
	(0.089)	(0.492)	(0.171)	(0.221)
$t = 3$	1.127***	0.050	0.922**	−0.447
	(0.248)	(0.712)	(0.342)	(0.257)
$t = 4$	0.871*	0.631	0.737	−0.222
	(0.457)	(0.728)	(0.483)	(0.317)
Average, years 0 to 4	0.912	0.099	0.714	−0.271
	(0.218)	(0.531)	(0.289)	(0.212)
Year 3 minus year 0	0.392	0.072	0.496	−0.397
	(0.138)	(0.503)	(0.165)	(0.155)
Pre-mandate coefficients zero, p	0.190	0.858	0.208	0.573

Notes: The table reports the coefficients of equation (8) plotted in Figure 3. The pre-mandate test is a Wald test that the three coefficients before the mandate are jointly zero. Standard errors in parentheses are clustered by state. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

References

Agarwal, Sumit, Shashwat Alok, Shiv Dixit, and Tejaswi Velayudhan, "Impact of Third-party Reporting on Corporate Tax Evasion: Evidence from Indian Tax Records," SSRN Working Paper 5029961 2024. Earlier version: Impact of the GST on Corporate Tax Evasion, ISB working paper, 2022.

Allingham, Michael G. and Agnar Sandmo, "Income Tax Evasion: A Theoretical Analysis," *Journal of Public Economics*, 1972, *1* (3-4), 323–338.

Badertscher, Brad A., Sharon P. Katz, and Sonja O. Rego, "The Separation of Ownership and Control and Corporate Tax Avoidance," *Journal of Accounting and Economics*, 2013, *56* (2–3), 228–250.

Basri, M. Chatib, Mayara Felix, Rema Hanna, and Benjamin A. Olken, "Tax Administration versus Tax Rates: Evidence from Corporate Taxation in Indonesia," *American Economic Review*, 2021, *111* (12), 3827–3871.

Bennedsen, Morten, Kasper Meisner Nielsen, Francisco Pérez-González, and Daniel Wolfenzon, "Inside the Family Firm: The Role of Families in Succession Decisions and Performance," *Quarterly Journal of Economics*, 2007, *122* (2), 647–691.

Bertrand, Marianne and Antoinette Schoar, "The Role of Family in Family Firms," *Journal of Economic Perspectives*, 2006, *20* (2), 73–96.

_ , Paras Mehta, and Sendhil Mullainathan, "Ferreting out tunneling: An application to Indian business groups," *The Quarterly Journal of Economics*, 2002, *117* (1), 121–148.

Besley, Timothy and Torsten Persson, "The Origins of State Capacity: Property Rights, Taxation, and Politics," *American Economic Review*, 2009, *99* (4), 1218–1244.

Bloom, Nicholas and John Van Reenen, "Measuring and Explaining Management Practices across Firms and Countries," *Quarterly Journal of Economics*, 2007, *122* (4), 1351–1408.

Burkart, Mike, Fausto Panunzi, and Andrei Shleifer, "Family Firms," *Journal of Finance*, 2003, *58* (5), 2167–2201.

Cameron, A. Colin, Jonah B. Gelbach, and Douglas L. Miller, "Bootstrap-Based Improvements for Inference with Clustered Errors," *Review of Economics and Statistics*, 2008, *90* (3), 414–427.

Chen, Shuping, Xia Chen, Qiang Cheng, and Terry Shevlin, "Are Family Firms More Tax Aggressive Than Non-family Firms?," *Journal of Financial Economics*, 2010, *95* (1), 41–61.

Choudhary, Keshav and Bhanu Gupta, "Dynamics of Evasion around Tax Thresholds: Evidence from Indian Firms," *Journal of Public Economics*, 2025, *252*, 105535.

Claessens, Stijn, Simeon Djankov, and Larry H. P. Lang, "The Separation of Ownership and Control in East Asian Corporations," *Journal of Financial Economics*, 2000, *58* (1–2), 81–112.

Crocker, Keith J. and Joel Slemrod, "Corporate Tax Evasion with Agency Costs," *Journal of Public Economics*, 2005, *89* (9-10), 1593–1610.

Desai, Mihir A., Alexander Dyck, and Luigi Zingales, "Theft and Taxes," *Journal of Financial Economics*, 2007, *84* (3), 591–623.

Directorate General of Central Excise Intelligence, "Central Excise Duty Evasion Detected, by Modus Operandi," DGCEI, Ministry of Finance, Government of India 2010.

Dyreng, Scott D., Michelle Hanlon, and Edward L. Maydew, "Long-Run Corporate Tax Avoidance," *The Accounting Review*, 2008, *83* (1), 61–82.

Edwards, Jeremy S. S. and Alfons J. Weichenrieder, "Ownership Concentration and Share Valuation," *German Economic Review*, 2004, *5* (2), 143–171.

Faccio, Mara and Larry H. P. Lang, "The Ultimate Ownership of Western European Corporations," *Journal of Financial Economics*, 2002, *65* (3), 365–395.

Goldberg, Pinelopi Koujianou, Amit Kumar Khandelwal, Nina Pavcnik, and Petia Topalova, "Imported Intermediate Inputs and Domestic Product Growth: Evidence from India," *Quarterly Journal of Economics*, 2010, *125* (4), 1727–1767.

Gopalan, Radhakrishnan, Vikram Nanda, and Amit Seru, "Affiliated Firms and Financial Support: Evidence from Indian Business Groups," *Journal of Financial Economics*, 2007, *86* (3), 759–795.

Gupta, Bhanu, Tejaswi Velayudhan, Vedant Vohra, and Ankur Garg, "Enhancing Revenue Collection Capacity through Automation: Evidence from a VAT," SSRN Working Paper 6433741 2026.

Hanlon, Michelle and Shane Heitzman, "A Review of Tax Research," *Journal of Accounting and Economics*, 2010, *50* (2–3), 127–178.

Jameson, Melvin, Andrew Prevost, and John Puthenpurackal, "Controlling Shareholders, Board Structure, and Firm Performance: Evidence from India," *Journal of Corporate Finance*, 2014, *27*, 1–20.

Kavya, B. and Santhakumar Shijin, "Ownership and Control of Widely and Closely Held Firms in India," *Vision: The Journal of Business Perspective*, 2017, *21* (4), 449–460.

Khelil, Imen and Hichem Khlif, "Tax Avoidance in Family Firms: A Literature Review," *Journal of Financial Crime*, 2023, *30* (4), 1021–1035.

Kleven, Henrik Jacobsen, Martin B. Knudsen, Claus Thustrup Kreiner, Søren Pedersen, and Emmanuel Saez, "Unwilling or Unable to Cheat? Evidence from a Tax Audit Experiment in Denmark," *Econometrica*, 2011, *79* (3), 651–692.

Kotsogiannis, Christos, Luca Salvadori, John Karangwa, and Innocente Murasi, "E-invoicing, Tax Audits and VAT Compliance," *Journal of Development Economics*, 2025, *172*, 103403.

Masulis, Ronald W., Peter K. Pham, and Jason Zein, "Family Business Groups around the World: Financing Advantages, Control Motivations, and Organizational Choices," *Review of Financial Studies*, 2011, *24* (11), 3556–3600.

Naritomi, Joana, "Consumers as Tax Auditors," *American Economic Review*, 2019, *109* (9), 3031–3072.

Okunogbe, Oyebola and Victor Pouliquen, "Technology, Taxation, and Corruption: Evidence from the Introduction of Electronic Tax Filing," *American Economic Journal: Economic Policy*, 2022, *14* (1), 341–372.

Pérez-González, Francisco, "Inherited Control and Firm Performance," *American Economic Review*, 2006, *96* (5), 1559–1588.

Pomeranz, Dina, "No Taxation without Information: Deterrence and Self-Enforcement in the Value Added Tax," *American Economic Review*, 2015, *105* (8), 2539–2569.

Porta, Rafael La, Florencio Lopez de Silanes, and Andrei Shleifer, "Corporate Ownership around the World," *Journal of Finance*, 1999, *54* (2), 471–517.

Porto, Edoardo Di, Marco Pagano, Vincenzo Pezone, Raffaele Saggio, and Fabiano Schivardi, "Careers and Wages in Family Firms: Evidence from Matched Employer-Employee Data," Working Paper 736, Centre for Studies in Economics and Finance (CSEF) 2024. ECGI Finance Working Paper No. 1022/2024.

Webb, Matthew D., "Reworking Wild Bootstrap-Based Inference for Clustered Errors," *Canadian Journal of Economics*, 2023, *56* (3), 839–858.